Yang Yan

Ph.D. Candidate, Zhejiang University / Westlake University

AI for Science | Computer Vision | Representation Learning | Structural Biology

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Research Profile

AI-for-science Ph.D. candidate building foundation models and open-source ML systems for cryo-electron microscopy. First author of a ***Nature Methods*** study introducing **Cryo-IEF**, the first foundation model for cryo-EM particle processing, pretrained on **65.3M particle images** and transferred to structural classification, pose-aware clustering, particle-quality ranking, and automated reconstruction. Current work extends this research through **CryoDECO**, a SOTA foundation-prior ab initio reconstruction framework for compositional and conformational heterogeneity. Longer term, my research aims to build structural world models and agentic reconstruction systems that reason over molecular identity, dynamics, and native cellular context.

Education

2023–Present **Ph.D. Candidate in Computer Science** — Zhejiang University / Westlake University
Joint Ph.D. program supervised by Prof. Fajie Yuan and Prof. Huaizong Shen. Research: foundation models for cryo-EM image processing, automated workflows, and heterogeneity analysis.

2020–2023 **M.S. in Translational Medicine (Engineering)** — Xiamen University
National Institute of Diagnostics and Vaccine Development in Infectious Diseases, supervised by Prof. Ningshao Xia. Research: machine learning for medical image analysis.

2016–2020 **B.S. in Electrical Information Engineering** — Northeastern University (Qinhuangdao)
Research included machine-learning applications in localization.

Selected Publications

A comprehensive foundation model for cryo-EM image processing 2026
Yang Yan, Shiqi Fan, Fajie Yuan, Huaizong Shen
Nature Methods, 23(1), 88–95. DOI: 10.1038/s41592-025-02916-8. Introduced Cryo-IEF as the first cryo-EM particle-processing foundation model and established a foundation-model paradigm for automated particle analysis.

Artificial intelligence foundation model automates cryo-EM structure determination 2026
Yang Yan, Huaizong Shen
Nature Methods Research Briefing, 23, 26–27, DOI: 10.1038/s41592-025-02917-7.

CryoDECO: Deconstructing Extreme Compositional and Conformational Heterogeneity in Cryo-EM via Foundation Model Priors 2026
Yang Yan, Yanwanyu Xi, Shiqi Fan, Yifei Wang, Ziyun Tang, Fajie Yuan, Huaizong Shen
LangTaoSha Preprint Server. DOI: 10.65215/LTSpreprints.2025.12.30.000075. SOTA foundation-prior model for heterogeneous cryo-EM reconstruction, advancing high-throughput panoramic structural biology.

Unsupervised cryo-EM images denoising and clustering based on deep convolutional autoencoder and k-means++ 2022
D. Zhang, **Yang Yan**, Y. Huang, B. Liu, Q. Zheng, J. Zhang, N. Xia
IEEE Transactions on Medical Imaging, 42(5), 1509–1521. Deep autoencoder and balanced K-means++ framework for cryo-EM denoising and clustering.

Machine learning for structure determination in single-particle cryo-electron microscopy: A systematic review 2021
J. G. Wu, **Yang Yan**, D. X. Zhang, B. W. Liu, Q. B. Zheng, X. L. Xie, S. Q. Liu, S. X. Ge, et al.
IEEE Transactions on Neural Networks and Learning Systems, 33(2), 452–472. Review of ML methods for single-particle cryo-EM structure determination.

Selected earlier medical-imaging publications 2022–2023
S. Li, B. Liu, S. Li, X. Zhu, **Yang Yan**, D. Zhang; D. Zhang, B. Liu, Y. Huang, **Yang Yan**, et al.
Complex & Intelligent Systems, 2022; Journal of Digital Imaging, 2023. Deep-learning and ordinal-regression methods for bone-age assessment.

Research Highlights

- **First cryo-EM foundation model:** introduced **Cryo-IEF**, the first foundation model developed for cryo-EM particle processing, pretrained on **∼65M particle images** and establishing a new paradigm for automated particle analysis in structural biology.
- **First fully automated reconstruction pipeline:** developed **CryoWizard**, a fully automated computational pipeline for single-particle cryo-EM reconstruction that streamlines raw data processing, particle ranking, and high-resolution 3D structure determination without manual intervention.
- **SOTA heterogeneous reconstruction:** developed **CryoDECO**, a state-of-the-art foundation-prior model for heterogeneous cryo-EM reconstruction; resolved **100-structure** simulated mixtures and advanced panoramic analysis of cellular machinery in native biological states.
- **Open-source Python package for cryo-EM data processing:** developed `cryodata`, a PyPI-installable data layer that converts CryoSPARC particle jobs into PyTorch-ready datasets with MRC/MRCS preprocessing, FFT/Hartley representations, LMDB storage, balanced sampling, and CryoSPARC-to-RELION metadata conversion.

Research Software and Open Source

2023–2026 **Cryo-IEF: First Foundation Model for Cryo-EM Particle Processing**
https://github.com/westlake-repl/Cryo-IEF
Designed and released the Cryo-IEF ecosystem, pretrained on ∼65M cryo-EM particle images. Built PyTorch inference and downstream tooling for CryoRanker and CryoClustering, enabling practical structural classification, pose clustering, and particle-quality assessment.

2024–2026 **CryoDECO: SOTA Heterogeneous Reconstruction with Foundation Model Priors**
https://github.com/yanyang1998/CryoDECO
Developed a prior-guided heterogeneous reconstruction framework that uses Cryo-IEF representations to reduce random ab initio initialization and disentangle compositional classification from 3D reconstruction.

2024–2025 **CryoWizard: Fully Automated Single-Particle Cryo-EM Reconstruction Pipeline**
https://github.com/SMART-StructBio-AI/CryoWizard
Built and extended an end-to-end computational pipeline integrating CryoRanker with CryoSPARC, streamlining raw movies, micrographs, or particles to high-resolution 3D volumes without manual intervention through command-line, web, and Chrome-extension interfaces.

2023–2026 **cryodata: Open-Source Python Package for Cryo-EM Data Processing**
https://github.com/SMART-StructBio-AI/cryoief-data
Created the **reusable data layer** used by Cryo-IEF, CryoDECO, and CryoWizard to convert raw CryoSPARC particle outputs into PyTorch-ready training and inference datasets. The package supports MRC/MRCS preprocessing, LMDB-backed dataset creation, Fourier/Hartley feature generation, balanced sampling, PyTorch data loading, and CryoSPARC-to-RELION metadata conversion. Packaged and documented the toolkit for `pip install cryodata`, turning lab-specific preprocessing into reproducible open-source scientific ML infrastructure.

Technical Skills

ML / Vision	Self-supervised learning, representation learning, foundation models, clustering, autoencoders, generative/reconstruction models, image denoising.
Structural Biology	Single-particle cryo-EM, cryo-ET, CryoSPARC/RELION workflows, ChimeraX visualization, MRC/MRCS data formats.
Systems	Multiprocessing data pipelines, parallel GPU training/inference.

Research Interests

My long-term research vision is to develop AI systems that understand molecular structure in its native, dynamic context. Building on the CryoDECO idea that foundation-model priors can break the circular dependency between classification and reconstruction, I am interested in next-generation structural world models that learn the latent laws of macromolecular identity, interaction, motion, and cellular context from large-scale cryo-EM/cryo-ET data. A central direction is adaptive manifold intelligence: models that infer the intrinsic degrees of freedom of a sample, choose appropriate latent capacity, and move beyond static structure determination toward continuous maps of biological state space. I am also interested in agent-based structural-biology systems that autonomously plan reconstruction strategies, diagnose failure modes, select data-processing actions, and connect imaging data with biochemical and multimodal biological evidence. The broader goal is panoramic structural biology: replacing single-target, purification-heavy workflows with AI-guided discovery of molecular machines, transient interactions, and conformational dynamics directly from complex native mixtures.